

Project Definition and Source Research

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1 Project Overview

Real Madrid Pulse is a data engineering pipeline designed for the collection, storage, and processing of news articles about Real Madrid published on `football-espana.net`. The system implements a **Medallion architecture** (Bronze / Silver / Gold) over a local Data Lake, orchestrated with Apache Airflow and deployed via Docker Compose.

The primary objective is to build an enriched text database that enables sentiment analysis and trend detection on media coverage of the club, aligned with the topics of the Data Engineering and Programming for Data Analysis courses.

2 Technical Architecture

2.1 Architecture Diagram

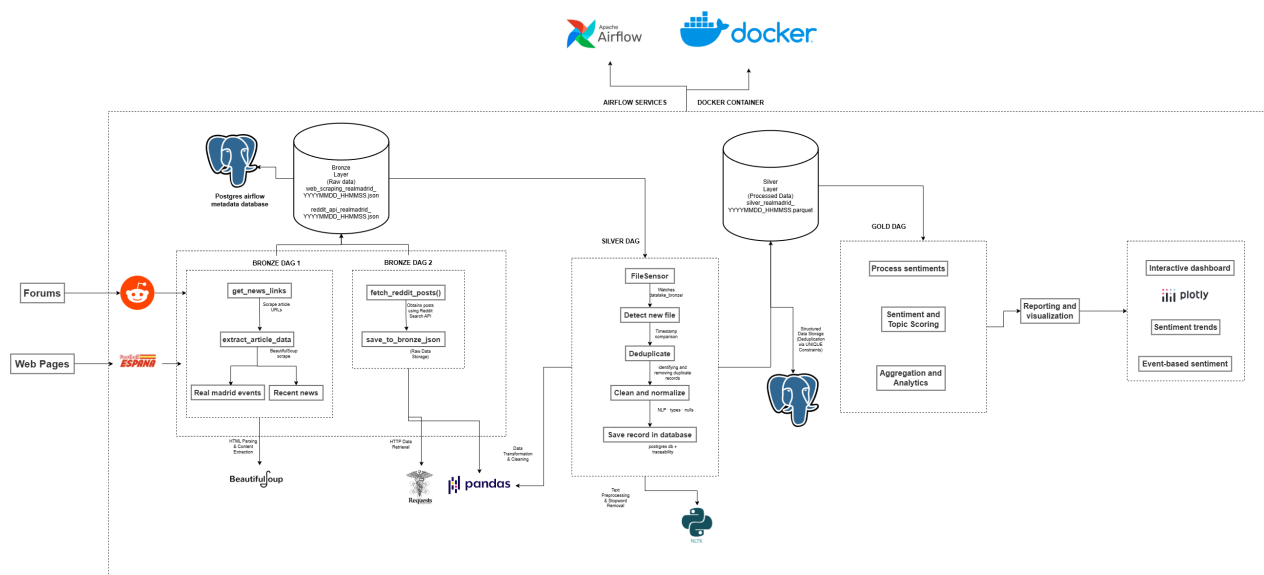


Figure 1: Architecture diagram

2.2 Overview

The system is designed as a data pipeline that follows a layered architecture composed of Bronze, Silver, and Gold layers. The pipeline is orchestrated using Apache Airflow and deployed within Docker containers to ensure scalability, reproducibility, and modularity.

The objective of this architecture is to extract data from external sources (forums and web pages), process and clean it, perform sentiment analysis, and finally provide insights through an interactive dashboard.

2.3 Architecture Components

2.3.1 Data Sources

The pipeline ingests data from two primary sources:

- **Forums (Reddit):** Data is retrieved using the Reddit Search API, focusing on posts related to Real Madrid.
- **Web Pages:** News articles are extracted from sports websites using web scraping techniques.

2.3.2 Orchestration Layer

- **Apache Airflow:** Responsible for orchestrating the entire pipeline through Directed Acyclic Graphs (DAGs).
- **Docker:** All services, including Airflow and dependencies, are containerized to ensure consistent execution environments.
- **PostgreSQL:** Used as Airflow's metadata database to store DAG execution states, logs, and scheduling information.

3 Bronze Layer (Raw Data)

The Bronze layer is responsible for ingesting and storing raw data without transformations.

3.1 Bronze DAG 1 (Web Scraping)

This DAG extracts data from web pages:

- `get_news_links`: Collects URLs of relevant news articles.
- `extract_article_data`: Extracts article content using BeautifulSoup.

The extracted data includes:

- Real Madrid events
- Recent news articles

3.2 Bronze DAG 2 (Reddit API)

This DAG handles data ingestion from Reddit:

- `fetch_reddit_posts()`: Retrieves posts using Reddit API.
- `save_to_bronze_json`: Stores raw data in JSON format.

3.3 Storage Format

Data is stored in the Bronze layer as raw JSON files using timestamp-based naming:

- web_scraping_realmadrid_YYYYMMDD_HHMMSS.json
- reddit_api_realmadrid_YYYYMMDD_HHMMSS.json

3.4 Tools Used

- BeautifulSoup (web scraping)
- Requests (HTTP calls)
- Pandas (basic structuring)

4 Silver Layer (Processed Data)

The Silver layer is responsible for cleaning, validating, and structuring the data. It writes processed data to **both** PostgreSQL and Parquet files in `datalake_silver/`, ensuring both structured queryability and file-based traceability.

4.1 Silver DAG

The Silver DAG performs the following steps:

- **FileSensor:** Monitors the Bronze layer to detect new files.
- **Detect New File:** Identifies newly ingested data based on timestamps.
- **Deduplication:** Removes duplicate records.
- **Cleaning and Normalization:**
 - Handling null values
 - Text normalization and basic cleaning
 - Standardizing formats and data types
- **Save to PostgreSQL:** Stores cleaned data into PostgreSQL for structured access.
- **Save to Parquet:** Persists a clean copy as a Parquet file in `datalake_silver/` for file-based traceability and future use.

4.2 Additional Processing

- Basic text cleaning (lowercasing, URL removal, punctuation removal, whitespace normalization)
- Advanced NLP tasks (tokenization, stopword removal, entity extraction) are delegated to the Gold layer

5 Gold Layer (Analytics – Planned)

The Gold layer focuses on generating business-level insights.

5.1 Gold DAG

- **Advanced NLP Processing:** Applies tokenization, stopword removal (NLTK), and entity extraction (spaCy) on Silver data.
- **Process Sentiments:** Applies sentiment analysis on processed data.
- **Sentiment and Topic Scoring:** Assigns polarity and extracts key topics.

- **Aggregation and Analytics:**
 - Sentiment trends over time
 - Event-based sentiment analysis

6 Visualization Layer (Planned)

The final layer provides insights through interactive dashboards.

- **Plotly Dashboard:**
 - Sentiment trends visualization
 - Event-based sentiment analysis

7 Data Flow

The pipeline follows a sequential flow:

1. Data is collected from Reddit and web pages.
2. Raw data is stored in the Bronze layer.
3. The Silver layer detects new data, performs cleaning and normalization, and stores results both in PostgreSQL and as Parquet files in `datalake_silver/`.
4. The Gold layer reads from PostgreSQL or Silver Parquet files, applies advanced NLP (spaCy) and sentiment analysis, and aggregates results.
5. Insights are visualized through dashboards.

8 Component Descriptions

The table below describes each component of the system: its purpose, the technology used, the inputs it receives, the outputs it produces, and its interactions with adjacent components within the pipeline.

Table 1: System Component Descriptions

Component	Purpose	Technology	Inputs	Outputs	Interactions
<code>bronze_ingestion_dag.py</code>	Scrapes raw HTML content from Real Madrid articles on <code>football-espana.net</code> and stores semi-structured data without transformation in JSON format.	Airflow 2.7.1, Python 3.8, requests, BeautifulSoup4, Pandas	Target URL category page on <code>football-espana.net</code>	JSON in <code>datalake_bronze/web_scraping_realmadrid_YYYYMMDD_HHMMSS.json</code> as	Feeds raw data to the Silver layer; triggered daily by the Airflow scheduler.

Component	Purpose	Technology	Inputs	Outputs	Interactions
reddit_api_bronze_ingestion_dag.py	Retrieves Real Madrid posts from the Reddit Search API and stores raw JSON data without transformation in the Bronze layer.	Airflow 2.7.1, Python, requests, Pandas	Reddit Search API endpoint	JSON datalake_bronze/reddit_api_realmadrid_YYYYMMDD_HHMMSS.json	in as Feeds raw Reddit data to the Silver layer; triggered daily by the Airflow scheduler.
silver_processing_dag.py	Detects new JSON files from the Bronze layer, performs data cleaning, normalization, and deduplication, and loads the structured data into both PostgreSQL and Parquet files in datalake_silver/.	Airflow 2.7.1, FileSensor, Python, Pandas, psycopg2	JSON files from datalake_bronze/	Structured records in PostgreSQL (realmadrid_db) and Parquet files in datalake_silver/	Consumes Bronze data; writes to PostgreSQL and to datalake_silver/ using Airflow connections.
PostgreSQL Metadata DB	Stores Airflow internal state: DAG runs, task instances, logs, variables, and connections.	PostgreSQL 13, psycopg2	Internal Airflow scheduler and webserver operations	Metadata queries for orchestration and monitoring	Used by Airflow services for orchestration and state management.
PostgreSQL Data Warehouse	Stores cleaned and structured data from the Silver layer for analytical and downstream processing purposes.	PostgreSQL 13, psycopg2	Processed data from Silver DAG	Queryable structured tables for analytics and Gold layer processing	Connected via Airflow using AIRFLOW_CONN_POSTGRES_REALMADRID.
datalake_bronze/	Raw data landing zone. Stores semi-structured JSON files preserving original extracted content without transformations.	Local filesystem, Docker bind-mount, JSON/UTF-8	Files generated by Bronze ingestion DAGs	JSON files available for Silver processing	Shared directory between host and Airflow containers.
datalake_silver/	Intermediate storage layer for processed data. The Silver DAG writes Parquet files here in addition to loading data into PostgreSQL, enabling file-based traceability and future ML pipelines.	Local filesystem, Docker bind-mount, Parquet/snappy	Processed data from Silver DAG	Parquet files: web_scraping_realmadrid_YYYYMMDD_HHMMSS.parquet, reddit_api_realmadrid_YYYYMMDD_HHMMSS.parquet	Shared directory between host and Airflow containers; consumed by Gold layer and ML pipelines.

9 Docker Compose Service Definitions

The `docker-compose.yml` file defines multiple services using Compose format version 3.7 (compatible with Docker Engine 18.06+). These services collectively enable orchestration, data storage, and pipeline execution.

Table 2: Docker Compose Service Definitions

Service	Base Image	Port	Key Variables	Function
postgres	postgres:13	5436:5432	POSTGRES_USER=airflow POSTGRES_PASSWORD=airflow POSTGRES_DB=airflow	Airflow metadata database. Stores DAG runs, task instances, logs, XComs, variables, and connections required for orchestration.
postgres-data	postgres:13	5437:5432	POSTGRES_USER=datauser POSTGRES_PASSWORD=datapass POSTGRES_DB=realmadrid_db	Analytical database. Stores cleaned and structured data generated by the Silver layer for querying and downstream analytics.
airflow-webserver	Custom (build: .) Base: airflow:2.7.1	8080:8080	EXECUTOR=LocalExecutor SQL_ALCHEMY_CONN=PostgreSQL (Airflow metadata) AIRFLOW_CONN_FS_DEFAULT	Airflow web interface. Allows monitoring DAG execution, viewing logs, and managing connections and variables. Initializes the database and creates the admin user at startup.
airflow-scheduler	Custom (build: .) Base: airflow:2.7.1	(none)	EXECUTOR=LocalExecutor SQL_ALCHEMY_CONN=PostgreSQL AIRFLOW_CONN_POSTGRES_REALMADRID	Core orchestration engine. Evaluates DAG definitions, schedules and executes tasks using LocalExecutor, and connects to both the metadata database and the analytical PostgreSQL database.

9.1 Volumes and Bind-Mounts

The project uses named volumes for database persistence and bind-mounts to share host directories with Airflow containers:

- `./airflow/dags` → `/opt/airflow/dags` (DAG definitions)
- `./datalake_bronze` → `/opt/airflow/datalake_bronze` (raw JSON data)
- `./datalake_silver` → `/opt/airflow/datalake_silver` (processed Parquet files)

Note: The current implementation may use absolute Windows paths for bind-mounts. While functional, it is recommended to migrate to relative paths (e.g., `./datalake_bronze`) for portability across environments.

9.2 Healthcheck and Dependencies

The `postgres` service includes a healthcheck based on `pg_isready` with a 5-second interval and 5 retries. The Airflow services declare `depends_on` with the `service_healthy` condition, ensuring that the metadata database is fully initialized before any Airflow component attempts to connect.

Additionally, the scheduler connects to the analytical database through the Airflow connection `AIRFLOW_CONN_POSTGRES_REALMADRID`, enabling the Silver layer to load processed data into PostgreSQL.

10 Technology Choice Justification

10.1 Apache Airflow 2.7.1 as the Orchestrator

Airflow is the de facto standard in data engineering for orchestrating ETL/ELT pipelines. The choice is justified by:

- **Declarative DAG definition in pure Python**, consistent with the course programming language.
- **The `@task` decorator and `.expand()` dynamic task mapping** allow article-level parallel extraction without additional concurrency code.
- **The `FileSensor`** implements the event-driven processing pattern: the Silver pipeline only activates when a new Bronze file exists, avoiding unnecessary processing.
- **Airflow Variables** act as a lightweight state registry (`last_processed_bronze_ts`), implementing idempotency without an additional database.
- **LocalExecutor** is sufficient for the current project volume and eliminates the operational complexity of a message broker such as Celery + Redis.

10.2 Pandas for Bronze → Silver Processing

Pandas is the core data processing library in the scientific Python ecosystem and is central to the Programming for Data Analysis course. Its use in the Silver layer is justified by:

- **Expressive API** for data cleaning: `dropna()`, `str.strip()`, `pd.to_datetime()`, etc., reducing transformation code volume.
- **Native integration with PyArrow** for writing Parquet via `df.to_parquet()`, without additional dependencies.
- **Adequate performance**: the data volume (~380 KB per file) is perfectly manageable in memory; distributed processing (Spark) is not required at this stage.
- **Alternative considered – Polars**: higher performance on large datasets but introduces an additional learning curve with no measurable advantage at the current project scale.

10.3 PostgreSQL for Silver Layer Persistence

The Silver layer persists processed data directly into a PostgreSQL relational database **and** writes Parquet files to `datalake_silver/`. This dual-write strategy provides both structured queryability and file-based traceability.

- **Structured storage and schema enforcement**: PostgreSQL enforces a well-defined schema (data types, constraints), ensuring data consistency and reducing errors during downstream processing.
- **Efficient querying capabilities**: SQL enables flexible and powerful querying (filtering, joins, aggregations), which is essential for analytical tasks such as sentiment analysis and trend detection.
- **Centralized data access**: Storing data in a database allows multiple components (e.g., Gold layer, dashboards, or ML models) to access the same consistent data source without relying solely on file-based pipelines.
- **Parquet as secondary output**: Parquet files in `datalake_silver/` provide columnar storage for ML workflows, historical reprocessing, and portability, following the naming convention defined in Section 11.
- **Integration with Airflow**: PostgreSQL is easily integrated into the pipeline using Airflow connections (e.g., `AIRFLOW_CONN_POSTGRES_REALMADRID`), enabling seamless data loading from the Silver DAG.
- **Transaction support**: ACID properties guarantee data integrity during inserts and updates, which is critical when processing incremental data.

10.4 BeautifulSoup4 + requests for Web Scraping

- **requests** handles HTTP communication with support for custom headers (User-Agent), timeouts, and error handling.
- **BeautifulSoup4** provides a high-level API for navigating the DOM tree and extracting text without raw regex over HTML.
- The **Universal extraction strategy** (all tags h1–h5, p, span, div, a, li, etc.) is consistent with the Bronze principle: capture everything and defer field selection to Silver.

10.5 Reddit Search API for Forum Data Ingestion

The Reddit Search API is used to retrieve Real Madrid related posts from the `r/realmadrid` and broader Reddit communities. Its use is justified by:

- **No authentication required** for read-only public post search, simplifying the ingestion setup.
- **Structured JSON responses** that map directly to analyzable fields (title, selftext, author, score, num_comments, created_utc).
- **Complementary perspective** to web scraping: Reddit data captures fan sentiment and community discussion rather than formal journalism.
- **Consistent with Bronze principles**: raw JSON is stored without transformation, preserving all API response fields for downstream selection in the Silver layer.

10.6 spaCy for Advanced NLP (Gold Layer)

Advanced NLP tasks including tokenization, lemmatization, named entity recognition, and stopword removal are handled by spaCy in the Gold layer. This deliberate separation ensures that the Silver layer remains a lightweight, fast cleaning step, while deeper linguistic processing is applied only when computing Gold-level features.

10.7 PostgreSQL 13 as the Metadata Store

PostgreSQL is the recommended production option over SQLite because it supports concurrent access from both the webserver and the scheduler without data corruption. PostgreSQL 13 is a stable LTS version with a well-maintained official Docker image, and its SQLAlchemy connection string (`postgresql+psycopg2://`) is native to Airflow.

10.8 Docker Compose for Deployment

Docker Compose allows the complete environment to be defined and launched with a single command (`docker compose up`), guaranteeing:

- **Reproducibility**: any team member can spin up an identical environment with the same versions of all dependencies.
- **Isolation**: services communicate over an internal Docker network; PostgreSQL is not exposed to the public network except on port 5436 for development.
- The **custom Dockerfile** extends the official Airflow image and installs project dependencies via pip, avoiding host-level installations.

11 File Nomenclature Conventions

The pipeline follows a standardized temporal naming convention for all files generated in the Bronze and Silver layers, guaranteeing natural chronological ordering (lexicographic = chronological), execution traceability, and prevention of name collisions in concurrent runs.

11.1 General Pattern

`<source_or_topic>_<YYYYMMDD>_<HHMMSS>.<extension>`

Where: YYYY = four-digit year; MM = month (01–12); DD = day (01–31); HH = hour in 24h format; MM = minutes; SS = seconds. The _ separator between date and time enables parsing with `re.search(r'(\d{8}_\d{6})', name)`.

11.2 Nomenclature Table by Layer

Table 3: File Naming Conventions by Data Lake Layer

Layer	Pattern	Real Example	Description
Bronze	web_scraping_realmadrid_YYYYMMDD_HHMMSS.json	web_scraping_realmadrid_20260409_000909.json	Raw HTML scraping data. Fixed prefix identifies source and topic.
Bronze	reddit_api_realmadrid_YYYYMMDD_HHMMSS.json	reddit_api_realmadrid_20260409_000909.json	Raw Reddit API data. Fixed prefix identifies API source.
Silver	web_scraping_realmadrid_YYYYMMDD_HHMMSS.parquet	web_scraping_realmadrid_20260409_000909.parquet	Clean structured data from web scraping. Same timestamp as source Bronze file for traceability.
Silver	reddit_api_realmadrid_YYYYMMDD_HHMMSS.parquet	reddit_api_realmadrid_20260409_000909.parquet	Clean structured data from Reddit API. Same timestamp as source Bronze file for traceability.
Gold (future)	sentiment_realmadrid_YYYYMMDD_HHMMSS.parquet	sentiment_realmadrid_20260409_000909.parquet	Sentiment model output. Prefix reflects the analysis applied.

11.3 Implementation in Code

The Bronze DAG generates the timestamp at the moment of saving the file:

```
1 timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
2 filename = f"web_scraping_realmadrid_{timestamp}.json"
3 full_path = os.path.join(BRONZE_PATH, filename)
4 df.to_json(full_path, orient='records', indent=2, force_ascii=False)
```

Listing 1: Bronze layer file naming – bronze_ingestion_dag.py

The Silver DAG extracts the timestamp from the filename for idempotency control:

```
1 def extract_ts_from_filename(filepath):
2     name = os.path.basename(filepath)
3     match = re.search(r'(\d{8}_\d{6})', name)
4     return match.group(1) if match else "00000000_000000"
5
6 # Idempotency check using Airflow Variable
```

```

7 last_processed_ts = Variable.get(
8     "last_processed_bronze_ts",
9     default_var="00000000-000000"
10 )
11 if latest_ts <= last_processed_ts:
12     raise ValueError(f"No new file. Last processed: {last_processed_ts}")

```

Listing 2: Timestamp extraction and idempotency check – silver_processing_dag.py

11.4 Test Files Excluded

Files with suffixes `_test`, `_test2`, `_test3`, `_test4` present in `datalake_bronze/` are empty development files. The Silver DAG regex `(\d{8}_\d{6})` automatically excludes them as they contain no valid timestamp.

12 End-to-End Data Flow

The following diagram describes the complete data flow from source to datalake:



Listing 3: Pipeline end-to-end data flow

13 Preprocessing Pipeline Definition

The preprocessing pipeline transforms raw semi-structured data collected in the Bronze layer into clean, structured, and analysis-ready data in the Silver layer. Given that the data originates from web scraping and Reddit API ingestion, it contains noise, missing values, and inconsistencies that must be addressed before any analytical or machine learning tasks. This pipeline performs a series of systematic transformations, including null handling, data type normalization, basic text cleaning, deduplication, and schema enforcement. Advanced NLP tasks are intentionally deferred to the Gold layer. All preprocessing steps are implemented within Apache Airflow DAGs, ensuring reproducibility, scalability, and automated execution of the data cleaning workflow.

13.1 Web Scraping Preprocessing Pipeline

The preprocessing pipeline for web-scraped data is implemented in the Silver layer using Apache Airflow and Python. Its objective is to transform raw semi-structured JSON data from the Bronze layer into clean, structured, and analysis-ready data stored in both PostgreSQL and Parquet files.

13.1.1 Null Handling

Several fields extracted from raw HTML may contain missing or incomplete values. The following strategies are applied:

- **Body Text:** Missing or empty article content is flagged using a boolean field `body_text_missing`, instead of being dropped, to preserve traceability.
- **Published Date:** If the primary timestamp (`published_at`) is missing, it is imputed using alternative fields such as `published_time`.
- **Reading Time:** Missing values are imputed using the median of the column to avoid bias from extreme values.
- **Author:** Missing authors are preserved as null and later cast to string for compatibility.

This approach ensures minimal data loss while maintaining analytical usability.

13.1.2 Data Type Normalization

The pipeline enforces consistent data types:

- `published_at` → datetime (UTC normalized)
- `reading_time_min` → integer (nullable)
- `body_text_missing` → boolean
- Text fields (`title`, `body_text`) → string

This normalization ensures compatibility with SQL queries and downstream analytics.

13.1.3 Text Preprocessing for Silver Layer

The Silver layer applies only basic text cleaning to prepare fields for downstream consumption:

- Lowercasing all text
- Removal of URLs
- Removal of punctuation and special characters
- Normalization of whitespace
- Noise filtering to remove irrelevant boilerplate phrases (e.g., comment forms, navigation text)

Two cleaned fields are generated:

- `body_text_clean`
- `title_clean`

Note: Advanced NLP operations — tokenization, stopword removal (NLTK), lemmatization, and named entity recognition (spaCy) — are **not** performed in the Silver layer. These tasks are the responsibility of the **Gold DAG**, which reads from the Silver outputs and applies the full NLP pipeline before sentiment scoring.

13.1.4 Outlier Detection and Handling

The pipeline applies basic statistical handling:

- **Reading Time:** Extreme or malformed values are filtered during parsing.
- Non-numeric values are converted to null and handled through median imputation.

Due to the nature of the dataset (textual web data), numeric outliers are limited, and aggressive filtering is avoided to preserve data integrity.

13.1.5 Deduplication

Duplicate records are handled at the database level:

- The `url` field is used as a unique identifier.
- SQL constraint: `ON CONFLICT (url) DO NOTHING`

This ensures that duplicate articles across ingestion batches are not inserted multiple times.

13.1.6 Schema Enforcement

The Silver layer enforces a structured schema before loading data into PostgreSQL:

- `url` (PRIMARY KEY)
- `title` (TEXT)
- `author` (TEXT)
- `published_at` (TIMESTAMP)
- `reading_time_min` (INTEGER)
- `body_text` (TEXT)
- `body_text_clean` (TEXT)
- `title_clean` (TEXT)
- `body_text_missing` (BOOLEAN)
- `bronze_source` (TEXT)

This schema is aligned with the target PostgreSQL table `articles_scraping`.

13.1.7 Design Justification

The preprocessing decisions are based on the characteristics of the collected web data:

- HTML extraction produces noisy and inconsistent text, requiring strong cleaning.
- Missing metadata (author, dates) is common and must be handled without discarding records.
- Basic text normalization in Silver ensures consistent input for downstream Gold NLP tasks.
- Dual output (PostgreSQL + Parquet) enables both structured querying and file-based ML workflows.
- Database storage enables structured querying and avoids duplication across ingestion batches.

13.2 Reddit API Preprocessing Pipeline

The Silver layer also processes raw JSON data ingested from the Reddit API. The preprocessing steps follow a similar structure to the web scraping pipeline, adapted to the Reddit data schema.

13.2.1 Key Fields Extracted

- `post_id` (PRIMARY KEY)
- `title` (TEXT)
- `selftext` (TEXT) – post body content
- `author` (TEXT)
- `subreddit` (TEXT)
- `score` (INTEGER)
- `num_comments` (INTEGER)
- `created_utc` (TIMESTAMP)
- `url` (TEXT)
- `selftext_clean` (TEXT) – basic cleaned version for Gold NLP
- `title_clean` (TEXT)
- `bronze_source` (TEXT)

13.2.2 Processing Steps

- **Null handling:** Empty `selftext` fields (deleted/removed posts) are flagged; missing authors preserved as null.
- **Data type normalization:** `created_utc` converted to UTC datetime; engagement metrics cast to integer.
- **Basic text cleaning:** Same cleaning steps as web scraping pipeline applied to `title` and `selftext`.
- **Deduplication:** ON CONFLICT (`post_id`) DO NOTHING at the database level.

14 DAG Documentation

14.1 Bronze DAG: Web Scraping

The Bronze DAG, identified as `football_scraping_universal_bronze`, is responsible for extracting raw data from web sources and storing it in the Bronze layer without transformations.

14.1.1 Schedule and Trigger

- **Schedule:** Daily (@daily)
- **Start Date:** April 1, 2026
- **Trigger Type:** Time-based scheduling via Airflow Scheduler

14.1.2 Tasks Description

- **get_news_links:**
 - Extracts article URLs from the Real Madrid category page.
 - Uses `requests` and `BeautifulSoup`.
- **extract_full_article_data:**
 - Performs deep extraction of HTML content for each article.
 - Captures multiple elements such as headings, paragraphs, metadata, links, images, and CSS classes.
 - Produces semi-structured records.
- **collect_results:**
 - Aggregates results from parallel extraction tasks.
- **save_to_bronze_json:**
 - Stores extracted data as JSON files in the Bronze layer.
 - File naming follows a timestamp-based convention.

14.1.3 Outputs

- JSON files stored in `datalake_bronze/`
- Naming format: `web_scraping_realmadrid_YYYYMMDD_HHMMSS.json`

14.1.4 Purpose

This DAG ensures that all raw data is captured in its original form, preserving maximum information for downstream processing and enabling reprocessing if needed.

14.2 Bronze DAG: Reddit API Ingestion

The Bronze DAG, identified as `reddit_api_bronze_ingestion`, is responsible for extracting data from the Reddit API and storing it in the Bronze layer in JSON format without transformations.

14.2.1 Schedule and Trigger

- **Schedule:** Daily (@daily)
- **Start Date:** April 1, 2026
- **Trigger Type:** Time-based scheduling via Airflow Scheduler

14.2.2 Tasks Description

- **fetch_reddit_posts:**
 - Sends HTTP requests to the Reddit Search API endpoint.
 - Uses query: "Real Madrid -Castilla" to filter relevant posts.
 - Retrieves up to 25 recent posts sorted by newest.
 - Extracts key attributes such as:
 - * title and content (selftext)
 - * author and subreddit
 - * engagement metrics (score, num_comments)
 - * timestamps and URLs
 - Adds ingestion metadata (retrieved_at).
- **save_to_bronze_json:**
 - Stores the extracted records as JSON files in the Bronze layer.
 - Uses timestamp-based file naming for versioning.

14.2.3 Outputs

- JSON files stored in `datalake_bronze/`
- Naming format: `reddit_api_realmadrid_YYYYMMDD_HHMMSS.json`

14.3 Silver Processing DAG: Web Scraping

The Silver DAG, identified as `football_silver_pipeline`, is responsible for detecting new web scraping data in the Bronze layer, performing data cleaning and transformation, and loading structured data into both PostgreSQL and Parquet files.

14.3.1 Schedule and Trigger

- **Schedule:** Daily (@daily)
- **Start Date:** January 1, 2024
- **Trigger Type:** Time-based scheduling with dependency on file availability via FileSensor

14.3.2 Tasks Description

- **wait_for_bronze_json (FileSensor):**
 - Monitors the Bronze directory for new JSON files.
 - Uses pattern matching: `web_scraping_realmadrid_*.json`
- **detect_new_file:**
 - Identifies the most recent Bronze file using timestamp extraction.
 - Compares against last processed file stored in Airflow Variables.
 - Ensures incremental processing.
- **clean_and_normalize:**

- Reads JSON data into a Pandas DataFrame.
- Extracts structured fields from raw metadata.
- Performs:
 - * Null handling and imputation
 - * Data type normalization
 - * Basic text cleaning (lowercasing, URL/punctuation removal, whitespace normalization)
 - * Noise filtering from HTML content
- Generates `body_text_clean` and `title_clean` fields.
- Produces a clean dataset ready for storage.
- **save_to_db:**
 - Inserts cleaned data into PostgreSQL using Airflow PostgresHook.
 - Uses `ON CONFLICT (url) DO NOTHING` for deduplication.
 - Adds metadata field `bronze_source`.
- **save_to_parquet:**
 - Writes the cleaned DataFrame as a Parquet file to `datalake_silver/`.
 - Uses the same timestamp as the source Bronze file for traceability.
 - Naming format: `web_scraping_realmadrid_YYYYMMDD_HHMMSS.parquet`

14.3.3 Outputs

- Structured records stored in PostgreSQL database `realmadrid_db`, table `articles_scraping`
- Parquet files stored in `datalake_silver/`

14.4 Silver Processing DAG: Reddit API

The Silver DAG for Reddit data, identified as `reddit_silver_pipeline`, processes raw Reddit JSON files from the Bronze layer and loads structured records into both PostgreSQL and Parquet files.

14.4.1 Schedule and Trigger

- **Schedule:** Daily (@daily)
- **Start Date:** April 1, 2026
- **Trigger Type:** Time-based scheduling with dependency on file availability via FileSensor

14.4.2 Tasks Description

- **wait_for_bronze_json (FileSensor):**
 - Monitors the Bronze directory for new Reddit JSON files.
 - Uses pattern matching: `reddit_api_realmadrid_*.json`
- **detect_new_file:**
 - Identifies the most recent Reddit Bronze file using timestamp extraction.
 - Compares against last processed timestamp stored in Airflow Variables.

- **clean_and_normalize:**
 - Reads JSON data into a Pandas DataFrame.
 - Handles null `selftext` values (deleted/removed posts).
 - Performs data type normalization (`created_utc` to UTC datetime, integer casting for metrics).
 - Applies basic text cleaning on `title` and `selftext`.
 - Generates `title_clean` and `selftext_clean` fields.
- **save_to_db:**
 - Inserts cleaned data into PostgreSQL using `ON CONFLICT (post_id) DO NOTHING`.
- **save_to_parquet:**
 - Writes the cleaned DataFrame to `datalake_silver/`.
 - Naming format: `reddit_api_realmadrid_YYYYMMDD_HHMMSS.parquet`

14.4.3 Outputs

- Structured records stored in PostgreSQL database `realmadrid_db`, table `reddit_posts`
- Parquet files stored in `datalake_silver/`

15 Results Evidence

This section presents visual evidence of the pipeline running successfully across all implemented components.

15.1 Bronze DAG: Web Scraping

Figure 2 shows the Bronze web scraping DAG `football_scraping_universal_bronze` executing successfully in Airflow, with tasks `get_news_links`, `extract_full_article_data` [20], `collect_results`, and `save_to_bronze_json` all completed.

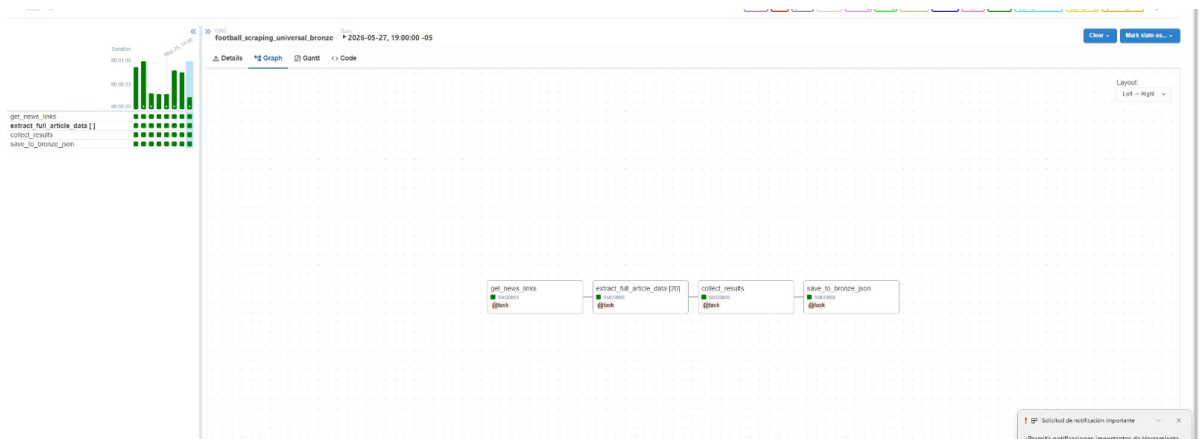


Figure 2: Bronze DAG `football_scraping_universal_bronze` – successful execution in Airflow.

15.2 Bronze DAG: Reddit API

Figure 3 shows the Bronze Reddit API ingestion DAG `reddit_api_bronze_ingestion` executing successfully, with tasks `fetch_reddit_posts` and `save_to_bronze_json` completed.

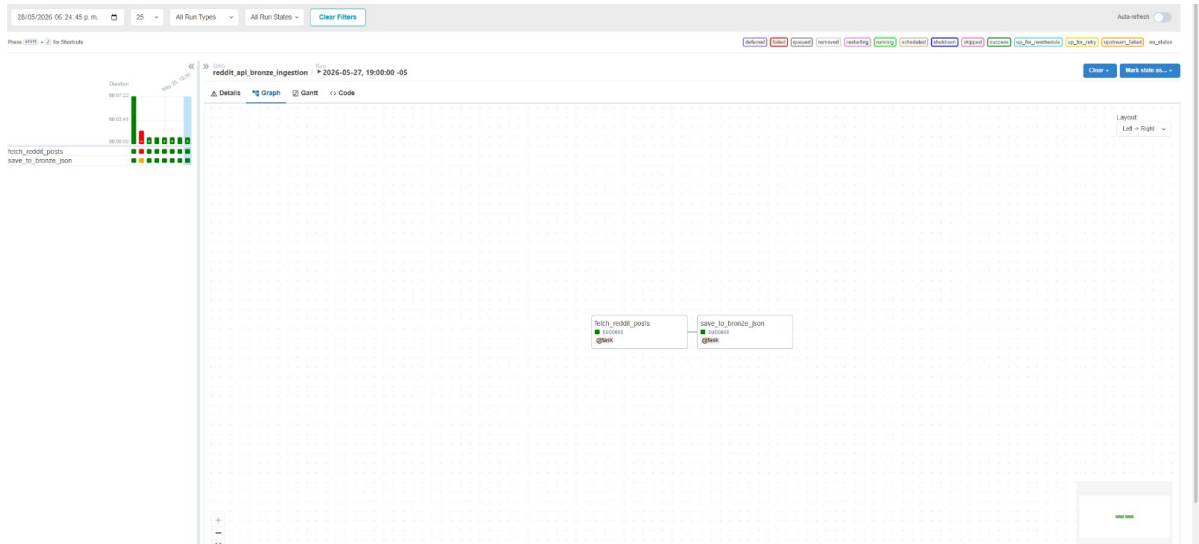


Figure 3: Bronze DAG `reddit_api_bronze_ingestion` – successful execution in Airflow.

15.3 Bronze JSON Sample: Web Scraping

Figure 4 shows the structure of a raw JSON file generated by the web scraping Bronze DAG, preserving all extracted HTML elements without transformation.

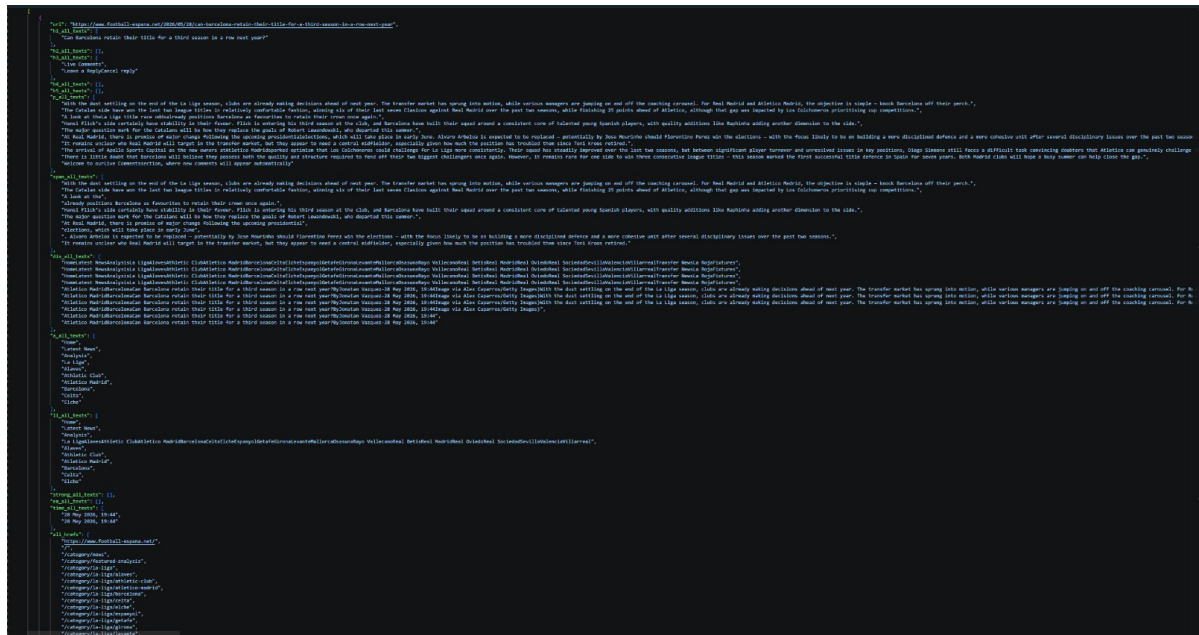


Figure 4: Raw JSON structure from the web scraping Bronze layer (`web_scraping_realmadrid_YYYYMMDD_HHMMSS.json`).

15.4 Bronze JSON Sample: Reddit API

Figure 5 shows the structure of a raw JSON file generated by the Reddit API Bronze DAG, with fields such as title, selftext, author, score, subreddit, and retrieved_at.

```
{
  "api_query": "Real Madrid -Castilla",
  "url": "https://www.reddit.com/r/realmadrid/comments/1tnebn/one_year_ago_today_xavi_alonso_was_announced_real/",
  "permalink": "https://www.reddit.com/r/realmadrid/comments/1tnebn/one_year_ago_today_xavi_alonso_was_announced_real/",
  "title": "One year ago today, Xavi Alonso was announced real Madrid manager.",
  "selftext": "",
  "author": "Significant.Fact6433",
  "score": 4,
  "num_comments": 2,
  "created_utc": 1779743086.0,
  "subreddit": "realmadrid",
  "retrieved_at": "2026-05-25T12:23:09.810053",
  "lang": "en"
},
{
  "api_query": "Real Madrid -Castilla",
  "url": "https://www.reddit.com/r/Aajainekhaya/comments/1tnebn/milk_maggie_i_think_i_cooked/",
  "permalink": "https://www.reddit.com/r/Aajainekhaya/comments/1tnebn/milk_maggie_i_think_i_cooked/",
  "title": "Milk Maggie I think I cooked 🍲",
  "selftext": "Milk Maggie in Kettle, with masala, butter, cheese, oregano and chilli flakes.",
  "author": "Novel.Huckleberry.77",
  "score": 1,
  "num_comments": 0,
  "created_utc": 1779743584.0,
  "subreddit": "Aajainekhaya",
  "retrieved_at": "2026-05-25T12:23:09.810053",
  "lang": "en"
},
{
  "api_query": "Real Madrid -Castilla",
  "url": "https://www.reddit.com/r/Northeastflames/comments/1tnebn/xabi_alonso_is_preparing_a_major_overhaul_at/",
  "permalink": "https://www.reddit.com/r/Northeastflames/comments/1tnebn/xabi_alonso_is_preparing_a_major_overhaul_at/",
  "title": "Xabi Alonso is preparing a MAJOR overhaul at Chelsea after failing to qualify for Europe.",
  "selftext": "When it looked as though Chelsea might finally be climbing out of the wreckage again, Stamford Bridge was forced to start over. Nothing remains of what Mauricio Pochettino and, above all, Enzo Maresca had begun to build. There is a re",
  "author": "Davy.warvy",
  "score": 4,
  "num_comments": 0,
  "created_utc": 1779743154.0,
  "subreddit": "Northeastflames",
  "retrieved_at": "2026-05-25T12:23:09.810053",
  "lang": "en"
}
```

Figure 5: Raw JSON structure from the Reddit API Bronze layer (reddit_api_realmadrid_YYYYMMDD_HHMMSS.json).

15.5 Silver DAG: Web Scraping Processing

Figure 6 shows the Silver DAG football_silver_pipeline executing successfully, with tasks wait_for_bronze_json, detect_new_files, clean_and_normalize [15], save_to_db [15], and update_last_processed all completed.

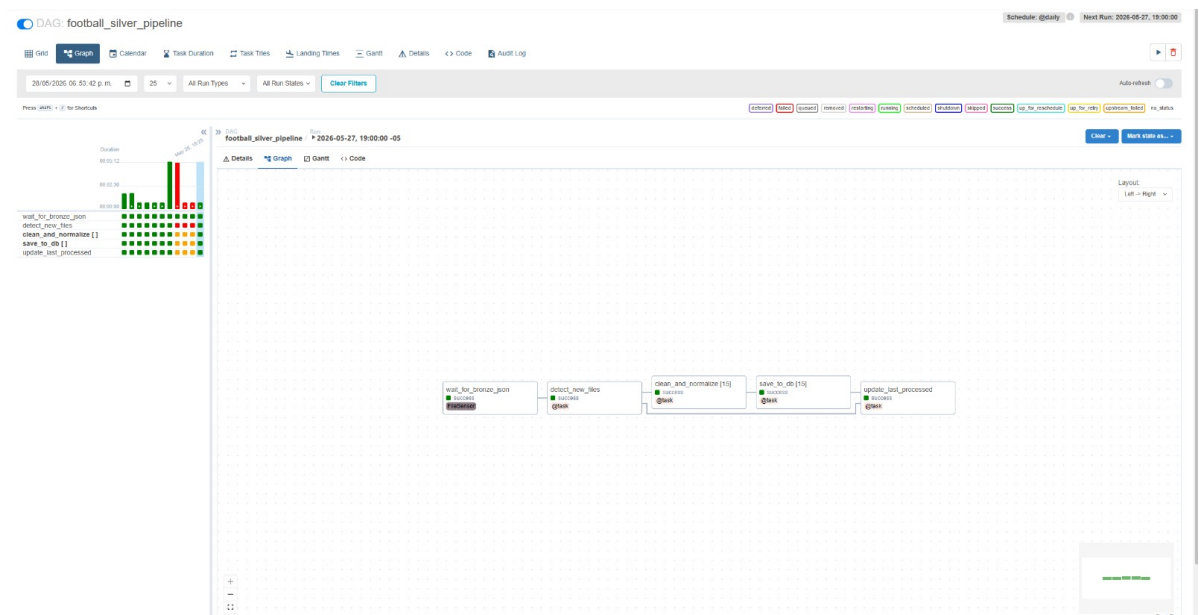


Figure 6: Silver DAG football_silver_pipeline – successful processing of web scraping Bronze data into PostgreSQL.

15.6 Silver DAG: Reddit API Processing

Figure 7 shows the Silver DAG `reddit_api_silver_pipeline` executing successfully, with tasks `wait_for_reddit_bronze_json`, `detect_new_file`, `clean_and_normalize`, and `save_to_db_and_parquet` completed.

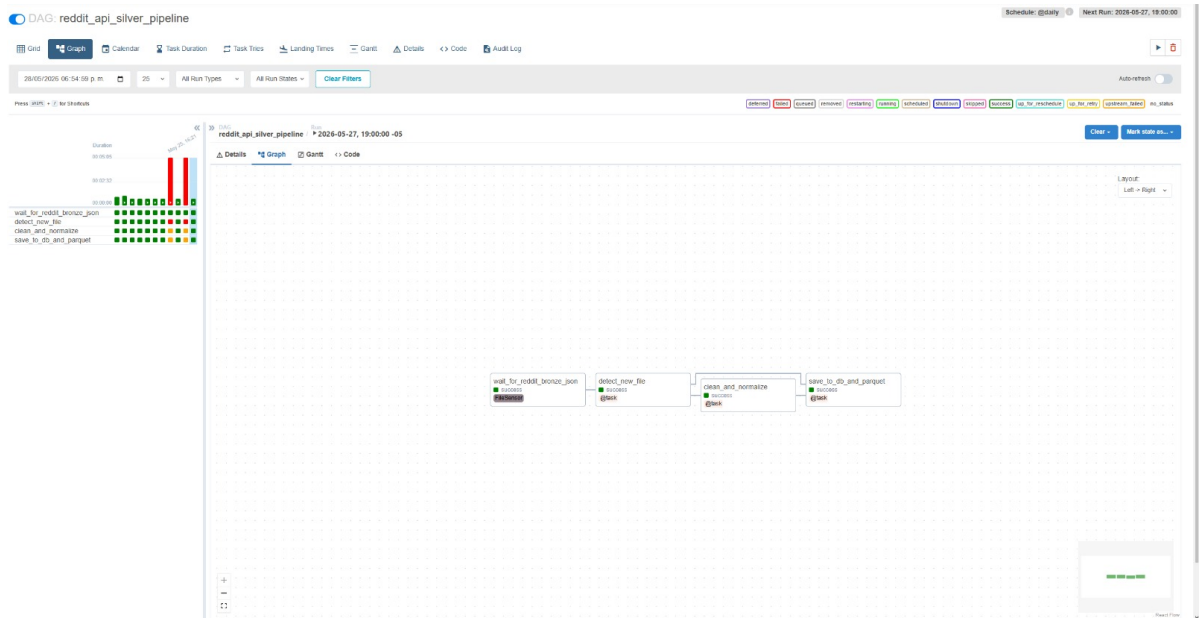


Figure 7: Silver DAG `reddit_api_silver_pipeline` – successful processing of Reddit API Bronze data into PostgreSQL.

15.7 Silver Output: Web Scraping – PostgreSQL Table

Figure 8 shows the structured records loaded into the PostgreSQL table `articles_scraping` after Silver processing of web scraping data, with fields `url`, `title`, `author`, `published_at`, `body_text`, `body_text_clean`, `title_clean`, `body_text_missing`, `source`, and `bronze_source`.

url	title	author	published_at	reading_time...	body_text	body_text_clean	title_clean	body_text_mis...	source	bronze_source
https://www.football...	Kylian Mbappe and a...	John Menzies	2026-05-25 13:49:22...	2	On Saturday, Real M...	on saturday real mad...	kylian mbappe and a...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid star inclu...	John Menzies	2026-05-25 13:19:22...	2	Spain head coach Lu...	spain head coach lu...	real madrid star inclu...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid star take...	Ruaridh Barlow	2026-05-25 10:03:07...	2	For the first time in h...	for the first time in h...	real madrid star take...	false	scraping	web_scraping_real...
https://www.football...	Spain manager on w...	Ruaridh Barlow	2026-05-25 07:19:27...	2	Spain manager Luis ...	spain manager luis d...	spain manager on w...	false	scraping	web_scraping_real...
https://www.football...	Lamine Yamal headlin...	Ruaridh Barlow	2026-05-25 05:55:46...	2	Spain manager Luis ...	spain manager luis d...	lamine yamal headlin...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid, Barcelo...	John Menzies	2026-05-25 00:21:24...	2	Real Madrid and Bar...	real madrid and barc...	real madrid barcelon...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid forced L...	John Menzies	2026-05-24 19:54:18...	2	It is taken for grante...	it is taken for grante...	real madrid forced to...	false	scraping	web_scraping_real...
https://www.football...	Bayern Munich respo...	John Menzies	2026-05-24 17:22:31...	2	Real Madrid are set f...	real madrid are set f...	bayern munich respo...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid star reac...	John Menzies	2026-05-24 16:56:34...	2	Real Madrid ended t...	real madrid ended th...	real madrid star reac...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid approve...	John Menzies	2026-05-24 15:47:15...	2	Real Madrid are abou...	real madrid are abou...	real madrid approve...	false	scraping	web_scraping_real...
https://www.football...	President confirms R...	John Menzies	2026-05-24 09:39:44...	2	Real Madrid are plan...	real madrid are plan...	president confirms r...	false	scraping	web_scraping_real...
https://www.football...	Dani Carvajal breaks ...	John Menzies	2026-05-23 18:57:50...	2	Dani Carvajal has no...	dani carvajal has no...	dani carvajal breaks ...	false	scraping	web_scraping_real...
https://www.football...	La Liga MD38 Round...	Ruaridh Barlow	2026-05-23 17:48:02...	2	A round-up of the ni...	a roundup of the nin...	la liga md38 roundu...	false	scraping	web_scraping_real...
https://www.football...	Real Madrid player r...	John Menzies	2026-05-23 17:00:43...	2	Real Madrid brought...	real madrid brought...	real madrid player ra...	false	scraping	web_scraping_real...
https://www.football...	Carvajal decide on s...	John Menzies	2026-05-23 16:19:00...	2	Real Madrid have en...	real madrid have end...	carvajal decide on s...	false	scraping	web_scraping_real...
https://www.football...	VIDEO: Mbappe retu...	John Menzies	2026-05-23 15:39:20...	2	Real Madrid are on c...	real madrid are on c...	video mbappe return...	false	scraping	web_scraping_real...
https://www.football...	VIDEO: Bellingham s...	John Menzies	2026-05-23 15:07:03...	2	Real Madrid are on c...	real madrid are on c...	video bellingham sc...	false	scraping	web_scraping_real...
https://www.football...	VIDEO: Carvajal assis...	John Menzies	2026-05-23 14:25:59...	2	Real Madrid are hopi...	real madrid are hopi...	video carvajal assis...	false	scraping	web_scraping_real...
https://www.football...	Jose Mourinho ready...	John Menzies	2026-05-23 09:29:57...	2	Real Madrid have a n...	real madrid have a n...	jose mourinho ready...	false	scraping	web_scraping_real...
https://www.football...	Jose Mourinho push...	John Menzies	2026-05-23 09:11:05...	2	Real Madrid have sta...	real madrid have star...	jose mourinho pushe...	false	scraping	web_scraping_real...

Figure 8: Silver layer output – `articles_scraping` table in PostgreSQL (`realmadrid_db`) populated from web scraping Bronze data.

15.8 Silver Output: Reddit API – PostgreSQL Table

Figure 9 shows the structured records loaded into the PostgreSQL table after Silver processing of Reddit API data, with fields including url, title, author, published_at, body_text_clean, title_clean, score, num_comments, subreddit, source, and bronze_source.

url	title	author	published_at	body_text	body_text_clean	title_clean	body_text_mis...	score	num_comments	subreddit	source	bronze_source
https://www.reddit.c...	One year ago today...	Significant-Fact5433	2026-05-25 16:19:46...	Milk Maggie in Kettl...	milk maggie in kettle...	one year ago today x...	true	4	2	realmadrid	reddit	reddit_api_realma...
https://www.reddit.c...	Milk Maggie I think I...	Novel_Huckleberry_77	2026-05-25 16:13:04...	Milk Maggie in Kettl...	milk maggie in kettle...	milk maggie i think i...	false	1	0	Aajmainekhaya	reddit	reddit_api_realma...
https://www.reddit.c...	Xabi Alonso is prepa...	Davy_wavy	2026-05-25 16:05:54...	**When it looked as ...	when it looked as th...	xabi alonso is prepari...	false	4	0	NorthwestBlues	reddit	reddit_api_realma...
https://www.reddit.c...	Aston Villa's Europea...	Comfortableglam	2026-05-25 16:04:31...	Aston Villa have reco...	aston villa have reco...	aston villas europe...	false	27	1	avfc	reddit	reddit_api_realma...
https://www.reddit.c...	What Parallel is this?	Rufia3034	2026-05-25 16:04:17...	What kind of parallel...	what kind of parallel...	what parallel is this	false	6	6	Panini	reddit	reddit_api_realma...
https://www.reddit.c...	UCL Mbappé	Skeleton1510	2026-05-25 15:59:39...	Is this normal????	is this normal	ucl mbappé	false	1	0	worldsoccerchamps...	reddit	reddit_api_realma...
https://www.reddit.c...	Official Florentino	Scary-Procedure-6018	2026-05-25 15:58:58...	Rest of the quote: "...	rest of the quote and...	official florentino	false	6	2	soccercirclejerk	reddit	reddit_api_realma...
https://www.reddit.c...	Enrique Riquelme: "I...	The_Big_Untalented	2026-05-25 15:55:34...	Rest of the quote: "...	rest of the quote and...	enrique riquelme i h...	true	17	19	soccer	reddit	reddit_api_realma...
https://www.reddit.c...	Enrique Riquelme: "I...	FootballInflamed	2026-05-25 15:47:31...	Source: https://baske...	source	enrique riquelme its ...	true	5	5	RealMadridFC	reddit	reddit_api_realma...
https://www.reddit.c...	Usman Garuba has su...	TheEastflamingGear	2026-05-25 15:44:17...	Source: https://baske...	source	usman garuba has su...	false	19	3	Euroleague	reddit	reddit_api_realma...
https://www.reddit.c...	Real Madrid signs M...	pica_fokes	2026-05-25 15:35:12...	By next week we will ...	by next week we will ...	real madrid signs mo...	true	46	22	soccer	reddit	reddit_api_realma...
https://www.reddit.c...	This guy should keep...	Artistic_Brother_310	2026-05-25 15:33:16...	By next week we will ...	by next week we will ...	this guy should keep...	false	17	36.25	RealMadridFC	reddit	reddit_api_realma...
https://www.reddit.c...	Real Madrid have pla...	FootballInflamed	2026-05-25 15:30:26...	Thank you all, truly ...	thank you all truly its...	real madrid have pla...	true	6	3	RealMadridFC	reddit	reddit_api_realma...
https://www.reddit.c...	Dani's speech before...	senormadridat0248	2026-05-25 15:23:41...	Thank you all, truly ...	thank you all truly its...	dani's speech before ...	true	35	1	realmadrid	reddit	reddit_api_realma...
https://www.reddit.c...	[Basketball] Real Ma...	Galego_ACB	2026-05-25 15:13:28...	**2027-2029** Gove...	2027/2029 governme...	basketball real madi...	true	9	1	realmadrid	reddit	reddit_api_realma...
https://www.reddit.c...	[SF] Preserve civiliz...	laughter_track	2026-05-25 15:07:00...	2027/2029 governme...	2027/2029 governme...	sf preserve civilization	false	1	1	shortstories	reddit	reddit_api_realma...
https://www.reddit.c...	Why Going Trophy...	liah_568	2026-05-25 14:50:30...	ever since this seaso...	ever since this seaso...	why going trophy...	false	1	2	RealMadridFC	reddit	reddit_api_realma...
https://www.reddit.c...	Carriero Asymmetric...	Tactical_Gaffer	2026-05-25 14:43:02...	&#x200B;Let's b...	amp lets be real for a...	carriero asymmetric...	false	0	0	TheOldZealand	reddit	reddit_api_realma...
https://www.reddit.c...	Manchester City is pl...	Peecece7896	2026-05-25 14:39:37...	Manchester city is pl...	manchester city is pl...	manchester city is pl...	true	1	0	SoccerNewsAndlum...	reddit	reddit_api_realma...
https://www.reddit.c...	Zero Real Madrid pla...	tw143d_m3n4t4	2026-05-25 14:21:53...	Manchester city is pl...	manchester city is pl...	zero real madrid play...	true	46	10	football	reddit	reddit_api_realma...

Figure 9: Silver layer output – Reddit posts table in PostgreSQL (realmadrid_db) populated from Reddit API Bronze data.

16 Current Status and Next Steps

16.1 Implemented Components

- Bronze Ingestion DAG (web scraping) fully functional with dynamic task mapping (.expand()).
- Bronze Ingestion DAG (Reddit API) fully functional with daily scheduling.
- JSON files in datalake_bronze/ with real Real Madrid article and Reddit post data.
- Silver DAG (web scraping) with new file detection, FileSensor, idempotency control, PostgreSQL loading, and Parquet output to datalake_silver/.
- Silver DAG (Reddit API) with equivalent processing pipeline for Reddit data.
- Docker Compose environment with PostgreSQL (metadata + analytical), Airflow Webserver, and Scheduler.

16.2 Pending Implementation

- **Gold layer:** application of advanced NLP (spaCy tokenization, NER, stopword removal) and sentiment model (NLTK VADER / scikit-learn) over Silver text data.
- **Gold DAG aggregation:** sentiment trends over time and event-based analysis.
- **Plotly Dashboard:** interactive visualization of sentiment trends and topic scoring.
- **Bind-mount path parameterization** for cross-platform portability (Windows / Linux / macOS).